

Sex, structural size, and pre-breeding isotope variation in Palmer Penguins

Abstract

We asked whether sex or structural size better explains pre-breeding red-blood-cell isotope variation in 344 Palmer Penguins. In the eligible baseline sample, males were 10% to 19.4% heavier than females across species. Information-theoretic model comparisons favored sex over size for nitrogen variation, while carbon patterns were species dependent: year led for Adelie and Chinstrap, and size plus year led for Gentoo. Seven focused one-at-a-time sensitivity analyses preserved the sex-over-size result for nitrogen in 21 of 21 candidate-set comparisons. These results distinguish robust sexual differentiation in trophic-position signals from more heterogeneous carbon-source patterns.

Introduction

Sexual size dimorphism may accompany ecological niche differentiation when males and females differ in prey or habitat use. Gorman et al. studied this question with structural measurements and carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) stable-isotope signatures in three congeneric Antarctic penguins ([doi:10.1371/journal.pone.0090081](https://doi.org/10.1371/journal.pone.0090081)). We reproduce their paper-informed candidate-model approach and ask a sharper comparative question: does sex or structural size receive more support as an explanation of isotope variation?

Data and provenance

The source comprises species-specific, citable Environmental Data Initiative packages for Adelie, Chinstrap, and Gentoo penguins. The normalized table retains all 344 observations, converts the source's sex code to missing, derives year from egg date, and assigns a stable species/sample row ID. Quality flags expose missing sex, morphology, and isotope measurements rather than silently deleting rows. The ASTRA manifest records each source, methodological decision, output, and evidence link; the Snakefile and Stencila runtime traces record how declared artifacts were produced.

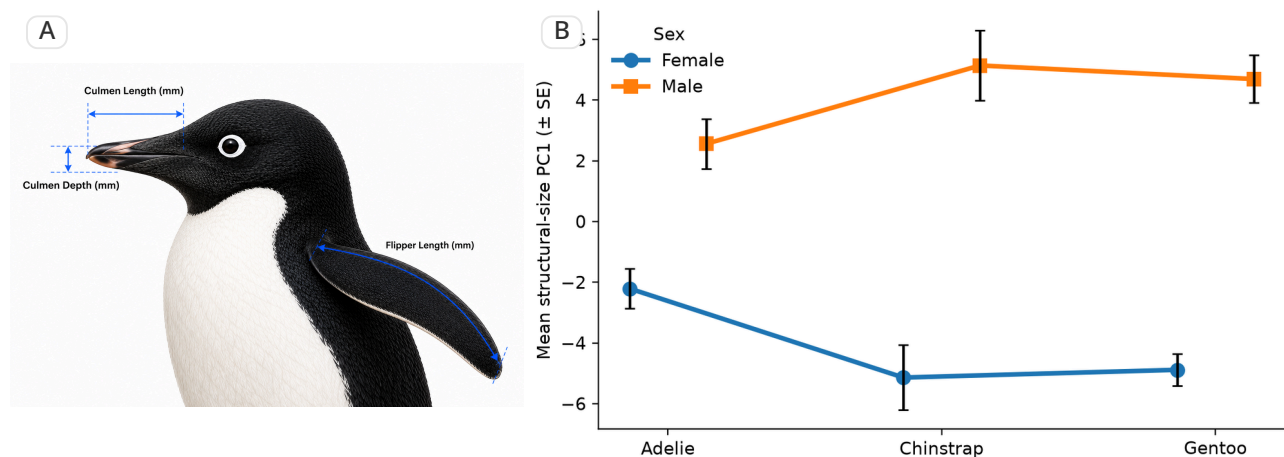


Figure 1: Morphological measurements and observed structural-size dimorphism. PC1 is oriented so larger scores correspond to longer flippers. **(A)** Measurement schematic for culmen and flipper dimensions. **(B)** Species-specific structural-size PC1 means and standard errors by sex.

Methods

The baseline restricts eligibility to birds recorded at nests that completed a clutch and uses endpoint-specific complete cases. Within each species, we center culmen length, culmen depth, and flipper length in their original units, extract PC1, and orient it to correlate positively with flipper length. Models are fit separately by species and isotope, with female and 2007 as reference levels.

The paper-like candidate set contains null, sex, size, year, sex + year, size + year, sex × year, and size × year models. We calculate small-sample-corrected AIC, ΔAICc , Akaike weights, and full unconditional model-averaged coefficients. For Adelie $\delta^{13}\text{C}$, the deterministic paper-like rule excludes stable row ade-119: the one observation with the largest absolute externally studentized residual across the two maximal interaction candidates.

Focused universes alter one decision at a time: inclusive eligibility, shared complete cases, standardized PC1, body mass, a joint additive sex-size-year model, retained outliers, or 2,000 resamples stratified by species, year, and sex with a fixed seed. This design probes identifiable choices without implying that every Cartesian combination is scientifically useful.

Results

Male mean body mass exceeded female mean body mass by 10% in Chinstrap, 16.7% in Gentoo, and 19.4% in Adelie penguins.

For $\delta^{15}\text{N}$, sex-containing models received substantially more baseline cumulative weight than size-containing models in all species. For $\delta^{13}\text{C}$, the best baseline model was year for Adelie (weight 0.45) and Chinstrap (weight 0.27), whereas Gentoo favored size + year (weight 0.69). Thus, sexual differentiation was clearest for nitrogen, while carbon evidence emphasized annual conditions or structural size depending on species.

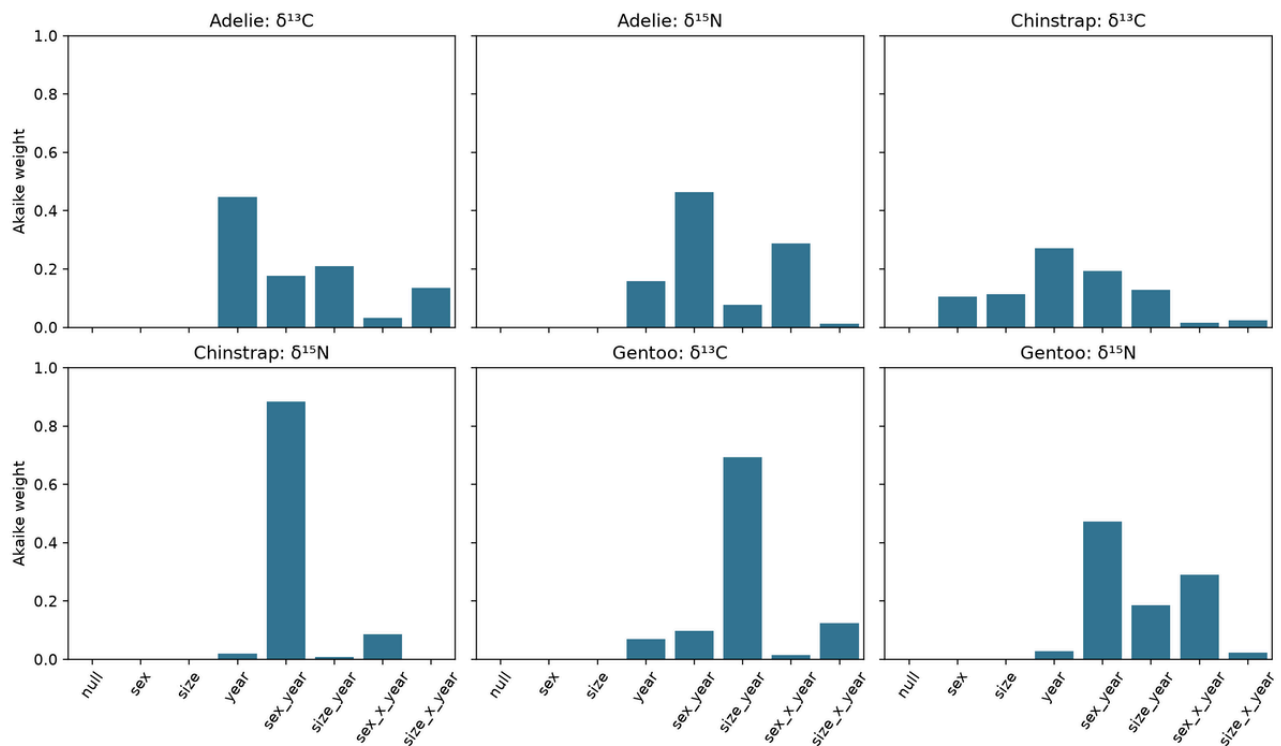


Figure: AICc weights for the eight baseline candidate models, separately by species and isotope.

Sensitivity analysis

Across the baseline and six alternative candidate-set universes, sex had more cumulative weight than size for $\delta^{15}\text{N}$ in all 21 species-universe comparisons. Gentoo $\delta^{13}\text{C}$ favored size over sex in 6 of 7 comparisons. Adelie and Chinstrap carbon evidence was less decisive, with the leading candidate or sex-size ordering changing under some choices. The joint model is shown separately because its single model necessarily contains both sex and size and therefore does not define competing cumulative weights.

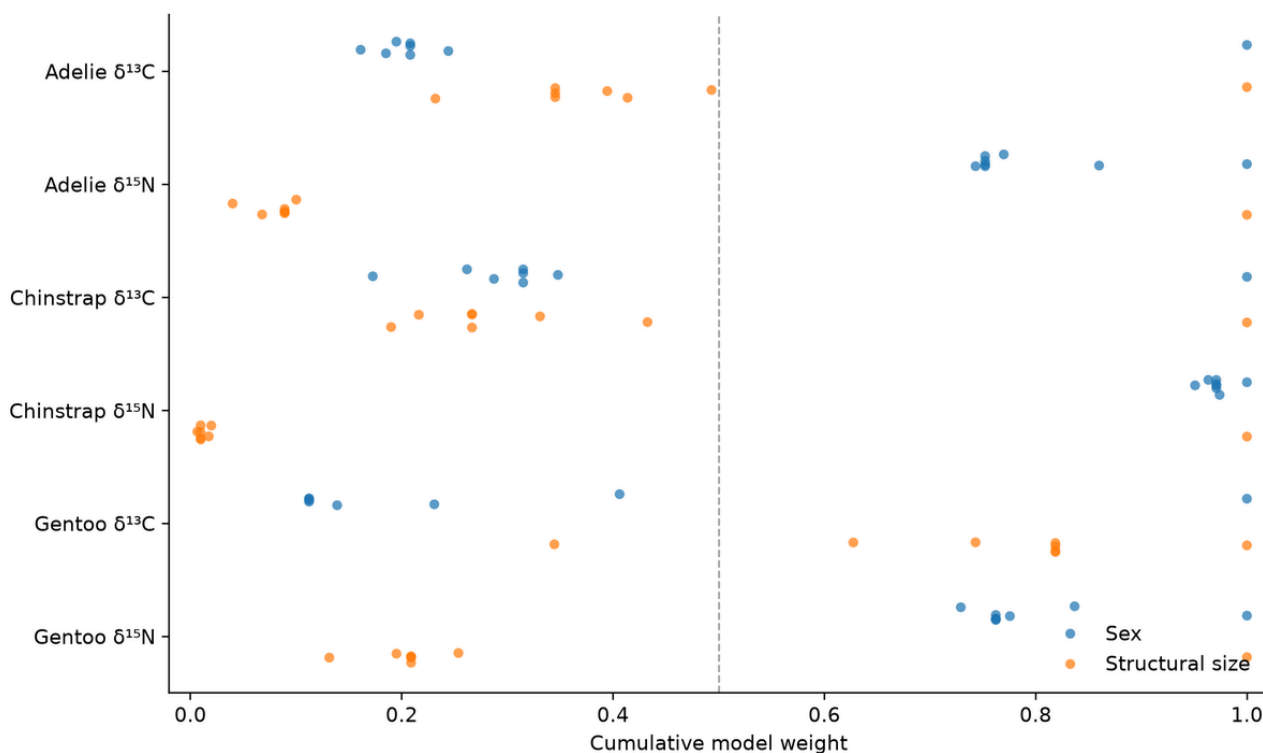


Figure: Cumulative Akaike weight assigned to models containing sex or structural size across the baseline and focused universes.

Limitations

Species and island are observationally confounded: all Gentoo records are from Biscoe and all Chinstrap records are from Dream, while Adelie records span three islands. Island effects therefore cannot be separated from species comparisons. Eligibility and missing-data rules reduce endpoint samples, isotope values integrate diet and habitat rather than directly measuring either, and AICc weights are relative to the declared candidate set. The residual exclusion operationalizes the published description reproducibly but cannot recreate an unreported visual judgment exactly.

Reproducibility

Python 3.13 dependencies are locked by `uv.lock`. Every Python-producing workflow rule runs through `stencila execute --trace`, the executable claims above read committed CSV artifacts, and the PDF is rendered with Stencila content credentials. `astra.yaml`, the eight universe files, the Snakefile, runtime evidence, and the credentialed PDF are merged into `graph.yaml` by the repository's current Stencila graph exporter.